

ABE Reviewer 2024

Volume 2

***Agricultural and Biosystems
Mechanization Planning,
Operation, Maintenance,
Management and Manufacturing***

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*Agricultural and Biosystems Mechanization
Planning, Operation, Maintenance,
Management and Manufacturing*

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ABE Licensure Exam Coverage

(Subjects, Topics and Subtopics are as specified in 2024 TOS)

Subject A	AGRICULTURAL AND BIOSYSTEMS POWER, ENERGY AND MACHINERY ENGINEERING AND ALLIED SUBJECTS (32%)
Topic I	Agricultural and Biosystems (AB) Power Engineering (4%)
Topic II	AB Mechanization Planning, Operation, Maintenance, Management and Manufacturing (8%)
Topic III	AB Machinery Specifications, Testing and Evaluation (3%)
Topic IV	AB Automation, Instrumentation and Control System (4%)
Topic V	Project Management, Feasibility Study Preparation/Evaluation, Research, Development and Extension, and Information System on AB Engineering (7%)
Topic VI	Laws, Professional Standards and Ethics (6%)
Subject B	LAND AND WATER RESOURCES ENGINEERING AND ALLIED SUBJECTS (32%)
Topic I	Hydrology (2%)
Topic II	Irrigation and Drainage Engineering (8%)
Topic III	Soil and Water Conservation Engineering (6%)
Topic IV	Aquaculture Engineering (2%)
Topic V	Fundamentals of Agricultural, Fishery, Ecological and Environmental Sciences (6%)
Topic VI	Mathematics and Basic Engineering (8%)
Subject C	AGRICULTURAL AND BIOSYSTEMS STRUCTURES, ENVIRONMENT ENGINEERING, BIOPROCESS ENGINEERING AND ALLIED SUBJECTS (36%)
Topic I	Agricultural Buildings and Structures (9%)
Topic II	Farm Electrification (4%)
Topic III	Environment Engineering (5%)
Topic IV	Agricultural and Bioprocess Engineering (14%)
Topic V	Food Engineering (4%)

Categories, their Descriptions & Percentages in the Exam

(Easy - 30%, Moderate - 50%, Difficult – 20%)

Easy1	Remembering. Recalling information, recognizing, listing, describing, retrieving, naming, finding
Easy2	Understanding. Explain, interpret, summarize, examine, paraphrase, classify
Moderate	Applying. Using the information in another familiar situation, implementing, carrying out, using, executing
Difficult1	Analyzing. Understanding, comparing, organizing, deconstructing, interrogating, finding
Difficult2	Evaluating. Hypothesizing, critiquing, experimenting, judging
Difficult3	Creating. Generating, designing, constructing, planning, producing, inventing

Instructions:

1. Select the best answer.
2. Shade only one answer per question in the answer sheet.
3. Don't make unnecessary marking in your answer sheet.

Questions:

1. The power source of a tractor.

- A. Engine
- B. Flywheel
- C. Power Take-Off (PTO)
- D. Accelerator

Answer: A

2. Prior to starting field operation, the tractor operator should check the adequacy of fuel, oil and _____.

- A. Air
- B. Battery fluid
- C. Water
- D. Engine lubricant

Answer: C

3. Three units of your company's hand tractors have just finished rotavating a dry 6-hectare mung bean field. The next task of the tractors is the plowing of a 5-hectare lowland rice field having a 20-cm average water depth. As the supervisor of tractor operation and maintenance, what priority task should you assign to the operators before starting the rice field operation?

- A. Change cage tires to rubber tires
- B. Change rubber tires to cage tires
- C. Change air filter
- D. Refill radiator with water

Answer: B

4. As the farm operations supervisor, which of these conditions should not be imposed to tractor operators in action?

- A. Free from any intoxication

- B. Good health condition
- C. Proper attire
- D. Tractor repair capability

Answer: D

5. Which plowing pattern is best if you want to minimize time loss by plowing immediately upon arrival of the tractor at the edge of the field?

- A. Circuitous pattern
- B. Land system pattern
- C. One way headland pattern
- D. Gathering headland pattern

Answer: A

6. The commonly used plowing pattern.

- A. Circuitous pattern
- B. Land system pattern
- C. One way headland pattern
- D. Gathering headland pattern

Answer: A

7. Which of the following is/are considered as secondary part(s) of a four-wheel tractor?

- A. Engine
- B. Controls and levers
- C. Chassis
- D. Parts for ensuring safety

Answer: C

8. The following four-wheel tractor controls and levers are commonly used and maintained, except _____.

- A. Stop/key button
- B. Brakes
- C. Clutch pedal
- D. Cantilever

Answer: D

9. Listed below are the commonly used and maintained power parts and hitches in four-wheel tractors, except _____.

- A. Three-point hitch
- B. Power Take-Off (PTO)
- C. Drawbar link
- D. Two-point hitch

Answer: D

10. As a planner, you want to greatly reduce time losses during plowing, harrowing, planting, weeding, spraying, harvesting and other mechanization operations in a contiguous 120-hectare area where land ownership is only half to three-fourths of a hectare per farmer. What is best to be proposed and implemented?

- A. Simultaneous land preparation
- B. Communal planting, weeding & harvesting
- C. Land consolidation
- D. Equitable land distribution

Answer: C

11. Maintenance and repair of tractors are important for longevity, economic efficiency and protection of the environment but the first and foremost reason is _____.

- A. Safety
- B. Preventing malfunction
- C. Retaining the tractor's value
- D. Increasing the tractor's life span

Answer: A

12. Your best guide in conducting maintenance is the _____.

- A. Manufacturer's advice
- B. Mechanic
- C. Parts supplier
- D. User's manual

Answer: D

13. A tractor maintenance routine that involves the checking of the engine oil level, coolant level, brakes and tire condition.

- A. General maintenance
- B. Regular inspection
- C. General inspection
- D. Preventive maintenance

Answer: B

14. To avoid tractor overloading which can cause early damage of components, the tractor should not continuously work at over ____% of its maximum engine power.

- A. 75
- B. 80
- C. 85
- D. 90

Answer: B

15. The time rate of doing work or delivering energy of a tractor, expressed as the amount of work done W or transferred energy divided by the time elapsed t or computed as W/t .

- A. Force
- B. Power
- C. Work
- D. Energy

Answer: B

16. What is the battery nominal voltage of a 4-wheel 90-hp tractor?

- A. 12 Vac
- B. 24 Vdc
- C. 9 Vdc
- D. 12 Vdc

Answer: D

17. The battery of a 24-volt combine harvester needs to be replaced. Two 12-volt batteries are to be used as its replacement. What should be the connection of the batteries?

- A. Series
- B. Parallel
- C. Series-Parallel
- D. Any connection

Answer: A

18. The 12-volt battery of a boom sprayer was drained. A 12-volt battery attached to a vehicle can be used for starting the engine. Jumper wires are available. What should be the connection of the two batteries to facilitate starting?

- A. Series
- B. Parallel
- C. Series-Parallel
- D. Any connection

Answer: B

19. A tractor cannot start due to a weak battery. It is being pushed by workers while lever is set to first gear. What should be foot-pressed and released abruptly to start engine?

- A. Brake
- B. Clutch
- C. Clutch and Brake
- D. Accelerator

Answer: B

20. To gradually move a tractor from stop position but with engine running, what should be released while pressing the accelerator?

- A. Brake
- B. Clutch
- C. Clutch and Brake
- D. Lever

Answer: B

21. To roll the tractor set at first gear in a flat land, what should be foot-pressed?

- A. Brake
- B. Clutch
- C. Clutch and Brake
- D. Accelerator

Answer: B

22. To slow down the tractor running at first gear in a flat land, what should be foot-pressed?

- A. Brake
- B. Clutch
- C. Clutch and Brake
- D. Accelerator

Answer: B (Clutch is better than clutch and Brake since purpose is only to slow down)

23. To gradually stop a fast running tractor, what should be foot-pressed?

- A. Brake
- B. Clutch
- C. Clutch and Brake
- D. Accelerator

Answer: C

24. What is the safest gear setting for a descending tractor in a 10% slope?

- A. Neutral
- B. Third gear
- C. Second gear
- D. First gear

Answer: D

25. In a routinary check before using the tractor, what should be checked first?

- A. Fuel
- B. Water
- C. Brake
- D. Oil

Answer: C

26. To roll a tractor from stop and neutral position with engine on in a flat land, what should be foot-pressed?

- A. Brake
- B. Accelerator
- C. Clutch and Brake
- D. Lever

Answer: B

27. The connection to the tractor of an erratic battery is to be stopped quickly. What battery terminal should be disconnected first?

- A. Positive
- B. Negative
- C. Both terminals
- D. Any terminal

Answer: A

28. A new battery is to be installed to a tractor. What battery terminal should be connected first?

- A. Positive
- B. Negative
- C. Both terminals
- D. Any terminal

Answer: A

Reference: Tractor Troubleshooting Video by PhilMech

29. The battery drains while tractor is not in use. What battery terminal should be disconnected when tractor is not in use?

- A. Positive
- B. Negative
- C. Both terminals
- D. Any terminal

Answer: A

30. What kind of hitch is best for a non-hydraulic disk plow?

- A. 1-point hitch
- B. 2-point hitch
- C. 3-point hitch
- D. 4-point hitch

Answer: C

31. To stabilize the 1-point fast hitch in a pull-behind disc harrow, a _____ hitching accessory is used.

- A. 3-point to 1-point hitch adapter
- B. 2-point to 1-point hitch adapter
- C. 3-point to 2-point hitch adapter
- D. PTO to drawbar lock key

Answer: A

32. What kind of hitch is best for a pull-behind disc harrow to facilitate fast hitching?

- A. 1-point fast hitch with 3-point to 1-point hitch adapter
- B. 2-point fast hitch
- C. 3-point fast hitch with 3-point to 1-point hitch adapter
- D. 4-point fast hitch

Answer: A

33. In a pull-behind 1-point fast hitch tandem disc harrow, what factor determines the depth of harrowing?

- A. Speed of harrowing
- B. Distance between the tandems
- C. Weight of harrow
- D. Offset angle

Answer: C

34. What kind of hitch is best for a hydraulic disc plow?

- A. 1-point hitch
- B. 2-point hitch
- C. 3-point hitch
- D. 4-point hitch

Answer: C

35. What kind of hitch is best for a non-hydraulic disc plow?

- A. 1-point hitch
- B. 2-point hitch
- C. 3-point hitch
- D. 4-point hitch

Answer: C

36. In a 3-point hitch hydraulic disc plow, which link or pair of links supports the hydraulic system?

- A. 1 upper link
- B. 2 lower links
- C. All 3 links
- D. Any of the 3 links

Answer: B

37. For best width of soil slice, the plow disc angle from the plowing direction should be _____.

- A. 15-25°
- B. 26-30°
- C. 40-45°
- D. 85-90°

Answer: C

38. For best depth of penetration, the plow tilt angle from the vertical line should be _____.

- A. 15-25°
- B. 26-30°
- C. 40-45°
- D. 85-90°

Answer: A

39. A 24-volt battery of a pineapple harvesting conveyor needs to be charged on field using 12-volt solar modules. What is the minimum number of solar modules required?

- A. One
- B. Two
- C. Three
- D. Four

Answer: B

Solution:

Minimum no. of modules = No. of solar modules in series
= $24\text{v}/12\text{v} = 2$ modules

40. A 24-volt battery of a boom sprayer needs to be charged on field using 6-volt solar modules with 2 modules in parallel to speed up charging. How many solar modules are required?

- A. Two
- B. Four
- C. Six
- D. Eight

Answer: D

Solution:

No. of solar modules in series = $24\text{v}/6\text{v} = 4$ modules

Total no. of solar modules = Modules in series x Modules in parallel = $4 \times 2 = 8$ modules

41. The old battery needs to be removed from a tractor. What battery terminal should be disconnected first?

- A. Positive
- B. Negative
- C. Both terminals
- D. Any terminal

Answer: B

Reference: Tractor Troubleshooting Video by PhilMech

42. What is the actual horsepower of a tractor needed to do 149.2 kw-hrs of work in 2 hours?

- A. 74.6 hp
- B. 57.5 hp
- C. 100 hp
- D. 105 hp

Answer: C

Solution:

$$\begin{aligned} P &= W/t \times \text{conversion factor} \\ &= 149.2 \text{ kw-hrs}/2\text{hrs} \times 1 \text{ hp}/0.746 \text{ kW} \\ &= 100 \text{ hp} \end{aligned}$$

43. In 4-wheel diesel tractor troubleshooting, what is the possible cause of hard starting engine and what is the corresponding solution?

- A. Weak battery, replace battery
- B. Low fuel, add fuel
- C. Old oil, change oil
- D. Defective spark plug, replace spark plug

Answer: A

44. The tractor engine overheats. What is the most possible cause and what is the corresponding solution?

- A. Hot weather, place tractor under shelter
- B. Problematic radiator, stop and cool down engine then check water level
- C. Broken piston ring, replace piston ring
- D. Low level of coolant, add coolant

Answer: B

45. The battery drains quickly while tractor is not in use. What is the most possible cause and what is the corresponding solution?

- A. Defective battery, replace battery
- B. Worn out battery plates, replace battery plates
- C. Battery is in partial short circuit condition, check and correct wiring from battery
- D. Low level of battery fluid, add battery solution

Answer: C

46. The tractor engine is running but not smoothly. What is the most possible cause and what is the corresponding solution?

- A. Clogged oil filter, clean oil filter
- B. Low water level in the radiator, add water
- C. Broken piston ring, replace piston ring
- D. Old engine oil, change oil

Answer: D

47. The engine of a hand tractor suddenly stops while the tractor is running. What is the most possible cause and what is the corresponding solution? The tractor is still under warranty.

- A. Defective engine, repair engine
- B. Broken sprocket weld, weld sprocket
- C. Broken shafting or chain, report immediately to the supplier
- D. Broken shafting or chain, have it repaired in a shop

Answer: C

48. The transmission of a hand tractor produces cracking sound when turning direction to left or right. What is the most possible cause and what is the corresponding solution? The tractor is no longer under warranty.

- A. Broken bearing, remove bearing cap and replace bearing
- B. Broken sprocket weld, weld sprocket
- C. Broken shafting or chain, report immediately to the supplier
- D. Broken shafting or chain, have it repaired in a shop

Answer: A

49. Leaking oil was observed in the hexagonal housing of a hand tractor. What is the most possible cause and what is the corresponding solution?

- A. Rust-damaged hexagonal housing, solder the hexagonal housing
- B. Loose bolt due to engine vibration, tighten bolt or replace loose thread bolt
- C. Loose bolt due to engine vibration, add washer and tighten bolt
- D. Oil used is of low viscosity, replace existing oil with higher viscosity oil

Answer: B

50. The rear part of a hand tractor was observed to be heavy when plowing or harrowing. What is the most possible cause and what is the corresponding solution?

- A. Unbalanced engine mounting, adjust the engine forward
- B. Unbalanced engine mounting, adjust the engine backward
- C. Unbalanced tractor assembly, increase counterweight in front of the tractor
- E. Unbalanced tractor assembly, lessen counterweight in front of the tractor
- F. in front of the tractor

Answer: A

51. A 100-hectare area needs to be disc harrowed and programmed to be finished by one tractor in 5 days at 8 hours per day operation. The tractor's harrowing speed is 7 km/hr while its field efficiency is assumed at 75%. If you are to design the harrow, what should be its width?

- A. 3 m
- B. 4 m
- C. 5 m
- D. 6 m

Answer: C

Solution:

$$\begin{aligned}\text{Width(m)} &= [\text{Area(ha)} \times 1000] / [\text{Speed(km/h)} \times \text{Time(hrs)} \\ &\quad \times \text{Field Efficiency(\%)}] \\ &= [100 \text{ has} \times 1000] / [7 \times (5 \times 8) \times 75] \\ &= 4.76 \text{ m}\end{aligned}$$

52. You need to purchase a tractor. Results of your calculation indicate that the PTO power needed for the routinary task intended for the tractor is 54 kW. What should be the minimum engine power in hp? Consider that the continuous work of the tractor at the PTO shaft should not exceed 78% of the engine power.

- A. 65 hp
- B. 90 hp
- C. 95 hp
- D. 100 hp

Answer: C

Solution:

Engine Power = [PTO Power (kW) / 0.78 x (kW to hp conversion)]

$$= 54 \text{ kW} / 0.78 \times (1 \text{ hp} / 0.746 \text{ kW})$$

$$= 92.8 \text{ hp minimum}$$

53. You are planning to acquire four-wheel tractor units for the primary tillage of consolidated farm having 1,000 hectares Area (A). Each tractor unit was computed to have a primary tillage field capacity (FC) of 1 hectare/hour. Primary tillage is limited to a working time (T) of 30 working days at 8 working hours/day including related non-tillage working hours. How many tractor units you should request for purchase?

- A. 2 units
- B. 3 units
- C. 4 units
- D. 5 units

Answer: D

Solution:

Approach: Dimensional/unit analysis

$$\text{Tractor Units} = [1 \text{ unit/FC}] \times [A/T]$$

$$= [1 \text{ unit} / (1 \text{ ha/hr})] \times [1,000 \text{ has} / (8 \text{ hrs/day} \times 30 \text{ days})]$$

$$= 4.17 \text{ units, rounded up to 5 tractor units}$$